Computer Architecture
Project Proposal



Project Title:

Multiplexer And Decoder Simulator

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Project Overview

The aim of this project is to assist students to learn how digital circuits work and how they convert inputs to outputs. It allows users to play with the circuits by mThis project is intended to make the students familiar with the digital circuits and their input-output behavior. Students can actually alter the input to a circuit by moving input switches and instantly observe the effect of their changes on the output(s) of the circuit.There are two circuits to experiment with: a 4-to-1 Multiplexer

Project Features

- 4-to-1 MUX: Four data inputs, two select switches, and one output.
- 2-to-4 Decoder: Two input switches and four output lamps.
- Truth Tables: Display the truth table of each circuit and highlight the current input row.
- Circuit Explanations: Provide a short explanation of how the MUX and decoder work.

Division of Work

Member 1 — Frontend **Syed Muhammad Ali Naqvi**	**Member 2 — Backend** **Syed Ali Raza Naqvi**
• Page design using HTML and CSS • Input switches and output lamps • Circuit pictures and truth-table display	• MUX and decoder logic • Truth-table generation • Preparation of 12 test cases

Tools and Technologies

HTML will use to create the page structure, CSS will design its layout, and JavaScript will perform the circuit calculations. All project files will be kept in three separate files: index.html, style.css, and script.js.

The application will be developed using HTML, CSS, and JavaScript only. It will run directly in a web browser without a database, server, or framework.

Expected Outcome

The simulator is expected to aid the understanding of these combinational circuits through practical experience. It will provide accurate results and visual responses for many combinations of inputs.

The final application will permit the user to interact with both the circuits, to understand their input-output behavior and to verify the results using truth tables.